REKORA

FROM IDEA TO PROTOTYPE.
FROM PROTOTYPE TO EVOLUTION.

PROJECT PROPOSAL

Integrated Engineering Project Development & Prototyping Ecosystem

SUBMITTED TO

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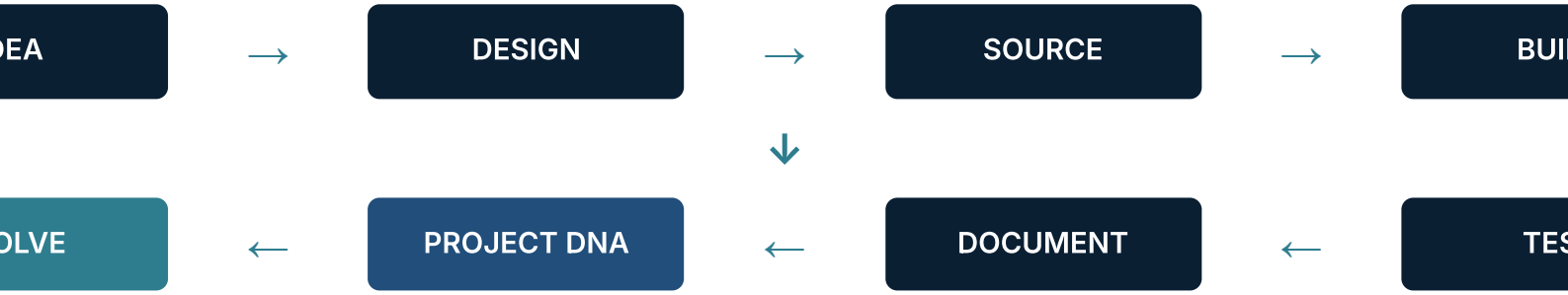
September 22, 2026

01 / EXECUTIVE SUMMARY

Engineering projects should not end at submission.

Today's student engineering projects involve a highly disconnected set of activities. A typical engineering student must independently navigate project discovery, component sourcing, CAD modification, fabrication, coding, testing, and documentation. Once a project is evaluated, the physical hardware is often discarded, and the engineering knowledge developed during the process is lost.

REKORA connects these fragmented activities into one continuous ecosystem.



> REKORA transforms an individual student project into a reusable, continuously evolving engineering asset.

To validate this model, we propose an initial controlled pilot at St. Joseph's College of Engineering. This pilot will introduce component rental to drastically lower student costs, direct campus-based 3D printing, and peer-to-peer resource exchange. We seek an initial seed allocation of ₹1,00,000–₹2,00,000 to establish the hardware inventory and deployment operations.

02 / THE PROBLEM

The academic engineering lifecycle is economically inefficient and structurally fragmented.

01

Fragmented Resources

Ideation, component sourcing, CAD modification, manufacturing, and documentation exist in isolated silos, causing immense friction for students.

02

Repeated Hardware Purchases

Students frequently purchase expensive components and hardware kits that are only required temporarily for a single prototype.

03

Limited Access to Fabrication

Moving from digital CAD to a physical prototype is a bottleneck. Students struggle with material selection and accessing 3D printing facilities.

04

Loss of Project Knowledge

Once a project is evaluated, the insights, troubleshooting steps, and design logic disappear. There is no centralized system to capture output.

05

Projects Restart From Zero

Because prior work is inaccessible, students cannot easily build upon previous engineering solutions.

06

Limited Continuity

Subsequent cohorts are forced to build identical basic systems year after year, preventing the organic evolution of ambitious engineering.

03 / THE OPPORTUNITY

The current inefficiency represents a substantial opportunity to redefine academic engineering.

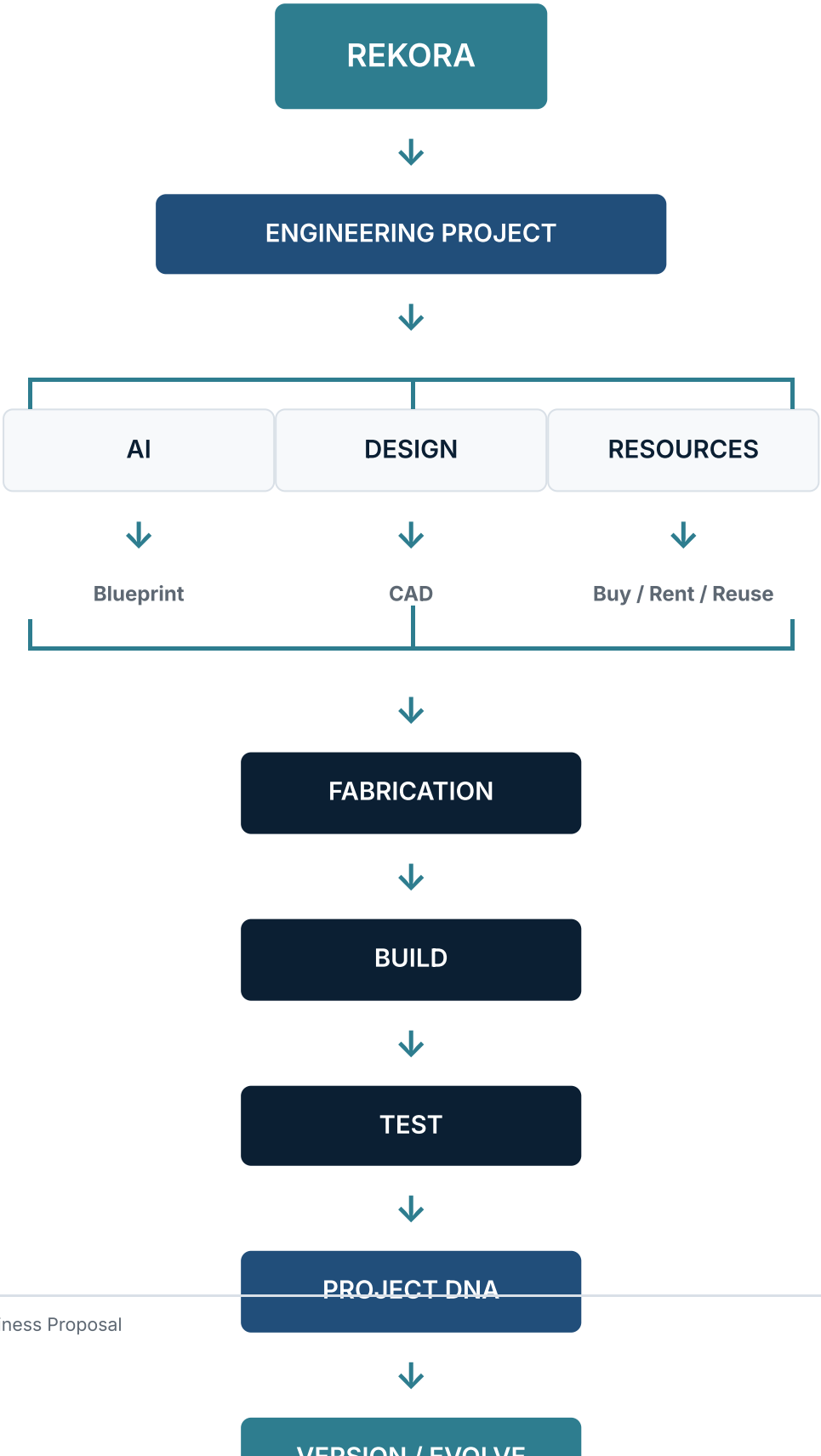
By consolidating the fragmented components of project development into a unified platform, we can reduce the financial barrier to innovation while dramatically improving the quality of student output.

There is an unprecedented opportunity to shift the paradigm from **"disposable student projects"** to **"cumulative engineering assets."** A platform that captures this value will not only serve students but will become an indispensable infrastructure layer for modern technical institutions.

> The integration of project discovery, hardware sourcing, and digital repositories opens the door for a circular academic economy—where resources are rented, reused, and continuously iterated upon.

04 / REKORA — THE INTEGRATED ECOSYSTEM

REKORA is designed to guide an engineering project from conceptualization to physical realization, and finally to archival and evolution.



05 / HOW REKORA WORKS

Discovery & AI

Users begin with a problem, budget, or components. The platform algorithmically recommends suitable initiatives, while the AI Project Engineer acts as a technical consultant to frame the constraints and generate a blueprint.

Component Sourcing

To reduce unnecessary purchases, REKORA integrates project BOMs directly with inventory availability. Students can choose to Buy, Rent, Use Existing, or Borrow components on an itemized basis.

CAD & Fabrication

REKORA connects digital design to the physical build. Students customize CAD models directly aligned with their requirements, then route them to campus 3D printing and manufacturing workflows.

Build & Documentation

During physical assembly, REKORA provides integration logic, troubleshooting, and standardized documentation frameworks to support academic submission rigor.

Project DNA

Every completed project receives a persistent digital identity preserving the exact configuration of hardware, software, and logic for future reference.

Evolution & Marketplace

Future users fork the repository to improve the physical design or code, while senior cohorts trade unused hardware via a peer-to-peer exchange.

06 / AI PROJECT ENGINEER

The REKORA AI Project Engineer is not a generic chatbot. It acts as a strict technical consultant integrated across the project lifecycle.

Project Consultation

Objective: Autonomous Agricultural Robot

Student: I want to build an autonomous agricultural robot under ₹5,000.

AI Engineer: Let's define the scope. What specific agricultural problem are you solving (e.g., seeding, weeding, monitoring)? What is your available timeline, and do you already possess basic components like microcontrollers or motor drivers?

Problem clarification
↓ **Budget**
↓ **Timeline & Skill level**
↓ **Available components**
↓ **Architecture & BOM**
↓ **Buy / Rent / Existing**
↓ **CAD requirements**
↓ **Testing plan**

↓

PROJECT BLUEPRINT

07 / PROJECT DISCOVERY & PROJECT MATCHER

REKORA allows users to discover projects through multiple strategic entry points rather than merely browsing a static list. The Project Matcher algorithmically recommends initiatives based on quantitative constraints.

By Problem Statement

"I want to solve localized water pollution."

By Available Components

"I have an ESP32, motors and an ESP32-CAM."

By Strict Budget

"I have exactly ₹5,000 to spend."

By Skill & Time

"I am a beginner and I have 30 days."

By Domain

"Robotics / IoT / Renewable Energy."

By Evolution

"I want to improve an existing v1.0 project."

> The platform ensures students undertake viable projects that precisely align with their financial capabilities and technical constraints.

08 / PROJECT PREVIEW → UNLOCK → FULL REPOSITORY

When a user discovers a project, they do not immediately receive the complete implementation logic. The platform utilizes a Preview & Unlock model.

AUTONOMOUS FLOATING WASTE COLLECTOR

Robotics • IoT • Computer Vision

Difficulty: Intermediate
Estimated Build: ₹4,500–₹6,000
Estimated Time: 2–4 Weeks

Technologies:
ESP32 | Computer Vision | GPS | Motor Control

ABSTRACT & PROBLEM

A low-cost autonomous surface vessel designed to identify and collect macro-plastics in static water bodies using localized computer vision and modular collection nets.

KEY CAPABILITIES

- Autonomous navigation
- Waste detection
- Remote monitoring
- Modular collection

WHAT YOU GET UPON UNLOCK

- Full Project Architecture
- CAD Resources & Source Code
- Complete BOM & Testing Plans

UNLOCK PROJECT REPOSITORY

The Preview Answers:

WHAT? WHY? WHO IS IT FOR? HOW DIFFICULT? WHAT DOES IT USE?

The Repository Answers:

HOW EXACTLY?

09 / PROJECT DNA & VERSION EVOLUTION

Every completed project can receive a persistent digital identity known as its **Project DNA**. This preserves the project objective, architecture, BOM, source code, CAD, testing results, and lessons learned.

```
PROJECT: Autonomous Waste Collector
│
├── v1.0 (Initial Prototype)
│
├── v1.1
│   └── Improved chassis design (Contributed by User A)
│
├── v1.2
│   └── Reduced BOM cost using alternate motors (Contributed by User B)
│
└── v2.0 (Major Evolution)
    ├── Integrated AI vision processing
    └── Full autonomous navigation capabilities
```

This version control philosophy allows future students to study, build, modify, test, and contribute to an existing project rather than discarding it. The project becomes a continuously evolving engineering asset.

10 / COMPONENT BUY / RENT / REUSE

REKORA connects project BOM items directly with campus inventory availability, significantly reducing the financial burden of hardware acquisition.

PROJECT BOM: SOURCING LOGIC

COMPONENT	STATUS	ACTION OPTIONS
ESP32 Microcontroller	Available in Inventory	**[BUY: ₹450]** or **[RENT: ₹80/wk]**
Motor Driver (L298N)	User already owns	**[USE EXISTING]**
Custom Sensor Housing	Digital CAD file only	**[FABRICATE / 3D PRINT]**
Standard Bearing 608ZZ	Consumable	**[BUY: ₹120]**

QR-Based Lifecycle Tracking

To support this ecosystem, physical components are managed via a QR-based lifecycle system (e.g., RK-ESP32-001). This operational infrastructure ensures quality control across the rental inventory.

dentity → **Project Usage** → **Return** → Inspe

11 / CAD → 3D PRINTING → FABRICATION

REKORA serves as the bridge between digital design logic and physical manufacturing realities, keeping manufacturing localized, efficient, and deeply integrated into the project lifecycle.



Current Capabilities

- 3D Printing (FDM / SLA)
- Custom Brackets & Housings
- Mechanical linkages
- CAD file modifications

Future Expansion

- Laser cutting
- CNC routing
- PCB fabrication routing
- External partner manufacturing

12 / RESOURCE MARKETPLACE & DOCUMENTATION

Senior / Student Resource Marketplace

To further the circular academic engineering economy, REKORA features a peer-to-peer exchange. This ensures that institutional resources remain active within the campus rather than sitting unused.

Seniors Can List:

- Unused project components
- Specialized tools
- Hardware kits
- Reference materials

Juniors Can:

- Buy directly at lower cost
- Rent for short-term projects
- Borrow specific tools
- Reuse proven hardware

Documentation Support

REKORA supports the academic rigor of project submission by providing structured documentation frameworks. The platform guides students in compiling necessary technical resources.

```
REKORA Documentation Framework
   ├── Abstract structure & Methodology
   ├── Auto-generated Block Diagrams
   ├── Circuit Schematics
   ├── Formatted Bill of Materials (BOM)
   ├── Testing & Validation logs
   └── Final Presentation structure
```

13 / BUSINESS & REVENUE MODEL

REKORA establishes a diversified, sustainable revenue architecture that minimizes reliance on low-margin commodity sales by monetizing services, access, and asset utilization.

Initial Revenue Model (Campus Pilot)

STREAM	DESCRIPTION
Project Repository Access	Users pay to unlock complete project resources, CAD, and source code where applicable.
Component Rental	Recurring high-margin utilization of physical inventory hardware.
Component Sales	Standard retail margin on consumable components and integrated project kits.
3D Printing & Fabrication	Manufacturing and service margins on direct campus-based 3D printing.
CAD & Design Services	Fees for paid design modifications and specific engineering assistance.
Marketplace Commission	Percentage commission on peer-to-peer transactions between students.

Future Revenue Opportunities (Scaling)

STREAM	DESCRIPTION
Technical Assistance	Premium troubleshooting, mentoring, and structured engineering support tiers.
Premium Platform Features	Subscription model for advanced AI Project Engineer capabilities.
Institutional Services	College-level SaaS subscriptions for centralized campus project archives.
Industry Sponsorships	Monetizing corporate-sponsored engineering challenges and prototype development.

14 / INITIAL COLLEGE PILOT

ST. JOSEPH'S COLLEGE OF ENGINEERING

Initial Controlled Pilot Environment

REKORA will not immediately pursue unstable nationwide deployment. Instead, we propose a controlled campus pilot. The objective of this college-first approach is to prove operational sustainability and establish willingness-to-pay before seeking external scale.

VALIDATE → GENERATE REVENUE → REINVEST

Pilot Validation Objectives

- Student demand and willingness to pay
- Component rental utilization rates
- Campus 3D printing demand
- Project repository engagement
- Efficacy of the AI Project Engineer
- Rate of resource reuse
- Marketplace transaction activity
- Inventory operational logistics

15 / 30-DAY CONTROLLED MVP

Because the core software platform is being developed internally by the founding team, we are prepared to execute rapidly. We propose launching a Minimum Viable Product (MVP) within 30 days of approval.

WEEK 1 — Platform Foundation

Deployment of the core database, student and admin interfaces, initial component catalogue, and basic project database infrastructure.

WEEK 2 — Intelligence & Sourcing

Activation of the AI Project Engineer, the algorithmic Project Matcher, project preview workflows, and Buy/Rent availability tracking systems.

WEEK 3 — Physical Pipeline

Establishment of the CAD and fabrication routing workflows, physical setup of the QR-based inventory tracking, and initiation of the senior resource marketplace.

WEEK 4 — Pilot Launch

Controlled onboarding of the first student cohort. Execution of pilot transactions, fabrication testing, feedback collection, refinement, and formal pilot launch.

16 / RESOURCE & BUDGET REQUIREMENTS

To execute the controlled campus pilot, we are seeking an initial seed requirement. Because software development is handled in-house, this capital will be deployed exclusively to establish the physical hardware ecosystem, inventory tracking, and fabrication capabilities required for the campus pilot.

> **INITIAL SEED REQUIREMENT: ₹1,00,000 – ₹2,00,000**

CAPITAL ALLOCATION AREA	ESTIMATED BUDGET RANGE (INR)
Initial Component Inventory Microcontrollers, sensors, basic robotics hardware for buy/rent	₹50,000 – ₹80,000
3D Printing & Fabrication Materials, maintenance of fabrication equipment	₹20,000 – ₹40,000
Contingency Reserve Operational buffer	₹10,000 – ₹20,000
QR & Inventory Tracking Labeling systems, physical storage infrastructure	₹5,000 – ₹10,000
Cloud Hosting & Technical Services Server costs, AI API overhead	₹5,000 – ₹10,000
Operations & Packaging Component handling, anti-static bags, rental logistics	₹5,000 – ₹10,000
Marketing & Campus Launch Student awareness campaigns, onboarding materials	₹5,000 – ₹10,000
Total Proposed Requirement Reference	**₹1,00,000 – ₹1,80,000**

* The proposed seed request spans ₹1,00,000–₹2,00,000 to retain necessary contingency and implementation flexibility during the pilot launch.

17 / RISK, RESPONSIBLE USE & ACADEMIC INTEGRITY

Risk & Mitigation Strategy

RISK FACTOR	MITIGATION APPROACH
Low component margins	Focus platform revenue heavily on integrated kits, high-margin rentals, and fabrication services rather than commodity trading.
Inventory capital lock-in	Start exclusively with highly standardized, high-utilization components in a strictly controlled inventory environment.
Hardware damage during rental	Implement QR lifecycle tracking, condition-check protocols, security deposits, and mandatory physical inspection upon return.
High technical support load	Leverage the AI Project Engineer for baseline troubleshooting, supplemented by comprehensive documentation and tiered support structures.
Intellectual Property issues	Establish clear project ownership terms, contributor attribution, and open-academic licensing records within the version history.
Operational complexity	Restrict the initial rollout entirely to a controlled college pilot environment to standardize workflows before any expansion.

Academic Integrity Boundary

> **REKORA DOES NOT SELL READY-MADE PROJECT SUBMISSIONS.**
> The platform strictly provides engineering resources, prototyping support, technical guidance, hardware access, and documentation frameworks. The student's actual implementation, testing, cognitive understanding, and academic responsibility remain entirely their own.

18 / INSTITUTIONAL BENEFITS

The successful deployment of REKORA creates systemic value across all tiers of the college environment, transitioning the institution from a consumer of hardware to an accumulator of engineering knowledge.

For the Institution

- Establishes a centralized, searchable digital archive of student innovation.
- Drastically improves the utilization of campus physical resources.
- Catalyzes a technology-driven student innovation ecosystem.
- Reduces redundant duplication of technical resources across departments.
- Demonstrates a commitment to modern project-based learning.

For Students

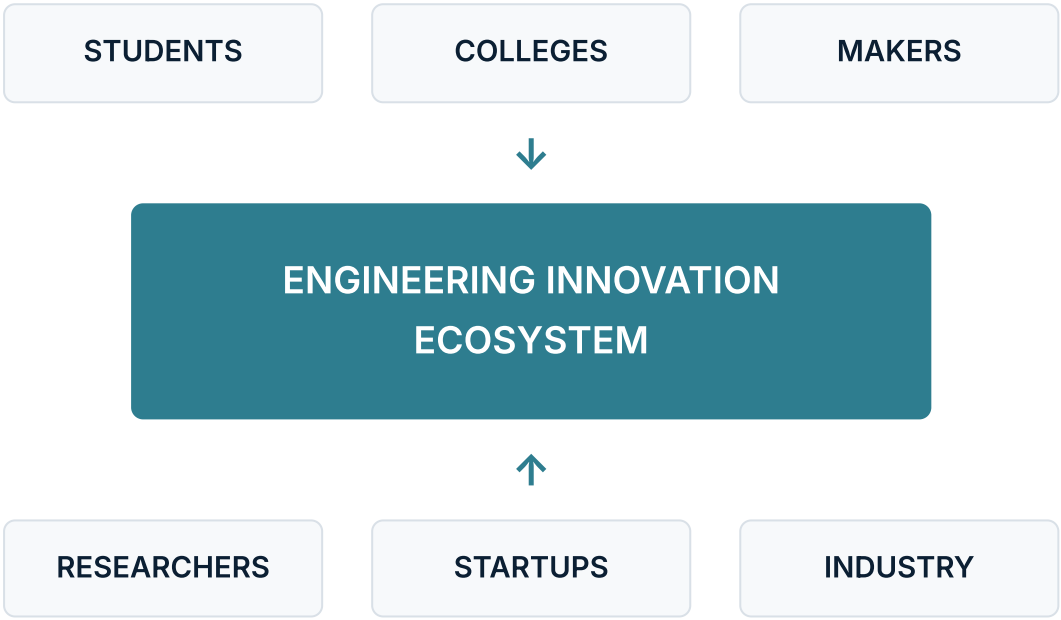
- Significantly lowers the hardware acquisition burden via rental capabilities.
- Provides immediate access to AI guidance, CAD, and fabrication tools.
- Unlocks the cumulative knowledge base of previous successful projects.
- Creates frictionless project discovery aligned with personal constraints.

For Faculty

- Grants unprecedented visibility into project evolution and student capabilities.
- Standardizes project records, blueprints, and version histories.
- Allows identification of recurring technical bottlenecks to improve curriculum.
- Streamlines the evaluation of authentic engineering work.

19 / LONG-TERM VISION

While execution begins with a tightly controlled college pilot, the long-term vision for REKORA is expansive. The platform is designed to evolve into an overarching engineering innovation ecosystem connecting multiple stakeholders.



Future Expansion Capabilities

- Multi-college project ecosystems
- Institutional SaaS project archives
- Advanced engineering marketplaces
- Broader decentralized fabrication networks
- Industry-sponsored problem statements
- Corporate prototype development
- Research collaboration platforms
- Engineering talent discovery networks

20 / FOUNDING TEAM & NEXT STEPS

REKORA is driven by a specialized team possessing the technical and operational expertise necessary to execute this infrastructure.

Reshma S

Founder & CEO

Overall vision, business strategy, and institutional partnerships.

Rogith Kumar R

Co-Founder & COO

Operations, execution, and continuous business process management.

NandhaKumar G

Co-Founder, Tech & Fab

Lead application development, 3D printing, and fabrication integration.

Monica S

Inventory Operations Manager

Component lifecycle, stock management, and QR procurement tracking.

Justin I

3D CAD & Product Design Lead

3D modeling, design modification, and prototyping architecture.

Sanjay B

Design & Client Relations Lead

Requirement gathering, student communications, and college relations.

Implementation Request

We seek formal institutional approval to initiate the 30-day MVP build and controlled pilot deployment. Upon approval and capital allocation, the founding team will immediately commence the establishment of the platform infrastructure, inventory procurement, and fabrication workflows at St. Joseph's College of Engineering.

REKORA

FROM IDEA TO PROTOTYPE.
FROM PROTOTYPE TO EVOLUTION.

"Today's project can become tomorrow's foundation."

REQUEST FOR INSTITUTIONAL APPROVAL
AND INITIAL SEED SUPPORT

₹1,00,000 – ₹2,00,000

St. Joseph's College of Engineering
Chennai